

Financial intermediaries reduce Transaction Costs

- Q: Why do we observe more indirect finance than direct finance (fact 3)?
- Financial intermediaries (e.g. mutual funds) make profits by reducing transactions costs for investors
 - Take advantage of economies of scale.
 - Bundling investors' funds together → Reduces transaction costs per individual investors.
 - Develop expertise to lower transactions costs.

Summary

Transaction Costs

- Transaction costs stop small savers and borrowers from direct financial investment.
- Financial intermediaries can help
 - economies of scale
 - develop expertise
- → Reduce transaction costs.

Asymmetric Information: Adverse Selection and Moral Hazard.

- Information dissemination——A key role in financial markets
- Asymmetric information occurs when one party to an economic transaction possesses **greater material knowledge** than the other party.
- E.g.
- CEO v.s. Shareholders (U.S.)
- Large Shareholders v.s. Small shareholders/ Retail investors (China)

Asymmetric Information leads to Adverse Selection and Moral Hazard

- Asymmetric information can take on many forms, and is quite complicated.
- **Asymmetric information leads to**
 - **Adverse selection**
 - **Moral hazard**

Asymmetric Information: Adverse Selection

- Adverse Selection
 1. Occurs when one party in a transaction has better information than the other party
 2. **Before** transaction occurs
 3. E.g. Hiding the information of a risky habit of sky diving when buying a life insurance.



Asymmetric Information: Adverse Selection

- E.g. Bank Loans and Adverse Selection
- Potential borrowers most likely to produce adverse outcome (high-risk borrowers) are ones most likely to seek loan and be selected.
- Why?
 - The bank **cannot perfectly distinguish** between high-risk and low-risk borrowers.
 - To protect itself from potential losses, the bank sets a high-interest rate.



Asymmetric Information: Moral Hazard

- Moral Hazard
 1. Occurs when one party has an incentive to behave differently once an agreement is made between parties
 2. **After** transaction occurs
 3. Hazard that borrower has incentives to engage in undesirable (immoral) activities, making it more likely that won't pay loan back.

Asymmetric Information: Moral Hazard

- E.g.: Moral hazard faced by banks
- Borrowers take excessive risks than they said when asking for money.
- Impact on bank...
 - Banks face higher default rates
- How to bank react?
 - Stricter lending policies or higher interest rates.
- Consequence? Better or worse development?
 - Borrowing becomes more expensive or difficult for other businesses.

Asymmetric Information: Moral Hazard

- Continue: Moral hazard faced by banks
- Why firms take excessive risks than they said?
- The bank expects the company to invest in safe, profitable projects (e.g. expand its business)
- Invest in high-risk, high-reward projects. e.g., risky stock trading, personal benefits.
- Why?
- Firms don't fully bear the consequences of failure.
 - If it succeeds, the company ...
 - makes huge profit. The bank does not share it.
 - If it fails, the company might default. But the bank ...
 - bears more losses than they supposed to bear.

Le Shi scandal: Asymmetric Information

- Asymmetric information occurs when one party to an economic transaction possesses **greater material knowledge** than the other party.



- **8 facts about financial structure**
 - Indirect finance is more important than direct finance
 - Banks are the most important source of external funds
- **Asymmetric information leads to**
 - Adverse selection (Occurs Before transaction)
 - Moral hazard (Occurs After transaction)

Results of adverse selection and moral hazard

- Adverse selection
 - Higher chances that a loan is made to a bad credit risk
 - **So, lenders decide not to make loans.**
- Moral hazard
 - Lowers the probability that the loan will be repaid
 - **So, lenders decide not to make loans.**

Asymmetric Information: Adverse Selection and Moral Hazard

- Recall: Asymmetric information leads to
 - Adverse selection
 - Moral hazard
- Use adverse selection and moral hazard to explain the 8 facts (financial structure).
- Fact # 6: Only large, well-established corporations have easy access to securities markets to finance their activities.

The Lemons Problem:

An example of How Adverse Selection Influences Financial Structure_(1 of 3)

- Lemons Problem in Used Cars.
- If we can't distinguish between “good” and “bad” (lemons) used cars, what price are we willing to pay for any used car?
 - An average of good and bad car values
- What is the result?
 - Good cars won't be sold, and the used car market will function inefficiently



Lemons Problem in Securities Markets

(2 of 3)

- If information about securities is not transparent
→ Lemons Problem
- We're willing to pay only average of good and bad securities' value.
- What type of cars will remain in the market?
- Result
 - Good securities undervalued and firms won't issue them;
 - Bad securities overvalued so too many issued
 - Many investors in this market in the long term?

The Lemons Problem:

An example of How Adverse Selection Influences Financial Structure (3 of 3)

- Lemons Problem in Securities Markets
- Only bad securities remain → Investors quit the market.
- Can market function well?
 - Adverse selection problem → Less developed financial market

How Adverse Selection Influences Financial Structure

- How to explain: Only large, well-established corporations have easy access to securities markets to finance their activities?
 - These firms provide investors more material and quality information about their firms.
 - So sellers agree to sell and buyers agree to buy.
 - i.e. Less adverse selection problem
- Eg. Why “New Third Board” is not functioning well?
 - Serving small and medium-sized enterprises
 - No. of listed firms 12,000→7,500
 - Limited disclosure and transparency
 - i.e. lack of reliable information



The Lemons Problem:

An example of How Adverse Selection Influences Financial Structure (3 of 3)

- Adverse Selection issue → Inactive market.
- How to solve?
 - Providing accurate and material information.
 - What type of financial firms is needed?

How to Solve Adverse Selection Problems

- If no asymmetric information issue → Problem solved.

1. Rating agency

- E.g. Rating for corporate bonds
- Private Production and Sale of **Information**.
- Firms sell unbiased information
- But, Free-rider problem
 - People enjoy a shared resource without having to contribute to it



How to Solve Adverse Selection Problems

1. Private Production and Sale of Information
 - Free-rider problem interferes with this solution
- Financial firms do not make money if having lots of free riders
- Who has the motivation to mitigate the problem of adverse selection, even it does not profit?
 - An efficient financial market is so important for the development of a country
- Why Fact 5: The financial system is among the most heavily regulated sectors of economy.

How to Solve Adverse Selection Problems

- Private firms cannot survive if having lots of free riders, but, government can.
- 2. Government regulation to require accurate Information (explains [Fact # 5](#))
 - E.g. annual audits of public corporations
 - Initiate regulations for financial industry
- Government solves all problem?
 - Expertise
 - Motivation
 - Salary
 - Bribery

How to Solve Adverse Selection Problems

- Financial intermediaries: experts that connect buyers and seller
 - Banks: find good firms and fairly price them
 - Key success compared with rating firms: they **holds mostly private loans** (debt financing outside public markets)
 - Avoid free-rider problem by making private loans (explains Fact # 3 and # 4)
 - More information is available for larger firms
 - Explains why large firms are more likely to use direct instead of indirect financing (fact #6)

How to Solve Adverse Selection Problems

4. Collateral and Net Worth

- Risk for debt holder: Default risk of borrowers.
- Collateral.
- $\text{Assets} > \text{Liabilities}$ (Net Worth).
- Explains Fact # 7

Summary of Adverse Selection

- Adverse Selection
 - Occurs when one party in a transaction has better information than the other party
 - An example of How Adverse Selection
 - The Lemons Problem
 - ❖ Information is not transparent
 - ❖ Result: Good ones are undervalued and only bad ones stay



Asymmetric Information: Adverse Selection and Moral Hazard

- **Asymmetric information leads to**
 - Adverse selection
 - **Moral hazard**

Moral Hazard

- Asymmetric information leads to
 - ✓ Adverse selection
 - Moral hazard
- Moral hazard
 - The seller's incentives to hide information and engage in activities against buyer's interest.
 - Before or after the transaction?
- How Moral Hazard Affects the Choice Between Debt and Equity Contracts ?
 - Has important consequences for whether a firm finds it easier to raise funds with debt than with equity (e.g. stock).

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Moral Hazard in Equity Contracts:
- the **Principal-Agent Problem**
- **The Principal:** The person or entity that delegates work or decision-making authority (e.g. shareholders or an employer).
- **The Agent:** The person or entity that performs the work or makes decisions on behalf of the principal (e.g., a company's CEO or an employee).
- The **core issue:** The agent has **more information** about their actions than the principal (**asymmetric information**) and may act in their **own self-interest** rather than maximizing the principal's benefit.

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- the **Principal-Agent Problem**
 - Result of **separation of ownership** by stockholders (*principals*) from control by managers (*agents*)
 - Managers act in own rather than stockholders' interest.
- Why interest misalignment?

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- CEOs' own interests
 - Cheat on the firms' profit, benefit him/her selves
 - e.g. Luxury life
 - Divert funds for personal preferences
 - Pet project; Risky investments
 - Power entrenchment
 - None profitable M&As
- Principal-Agent Problem
 - Caused by Interest misalignment
 - Is an example of moral hazard

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Tools to Help Solve the Principal-Agent Problem
 1. Production of Information: Monitoring.
 - Check on what is going on about managers and firms
 - Costly state verification.
 - If having multiple shareholders, free-rider problem;
 - Making it hard for firms to issue common stock to raise capital (Explain fact # 1).

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Tools to Help Solve the Principal-Agent Problem
 1. Production of Information: Monitoring
 2. Government Regulation to Increase Information
 - Governments have laws to force firms to behavior ethically.
 - What's the problem?
 - Increased Compliance costs for firms
 - Bureaucratic Inefficiencies
 - Enforcement challenges
 - Still find ways to hide information or manipulate reports since they're insiders

How Moral Hazard Affects the Choice Between Debt and Equity Contracts

- Tools to Help Solve the Principal-Agent Problem
 1. Production of Information: Monitoring
 2. Government Regulation to Increase Information
 3. Financial Intermediation
 - Financial intermediaries have the ability to avoid the free-rider problem (Explain fact 3).
 - e.g. Venture capital.
 - Keep a close watch on the firm's activities
 - Enjoy all gains
 - High-tech firms.

Summary

- Basic Facts About Financial Structure:
 - We reviewed 8 basic facts concerning the structure of the financial system
- Transaction Costs:
 - How transaction costs can hinder capital flow.
 - The role financial institutions play in reducing transaction costs

Summary

- Asymmetric Information leads to
 1. Adverse Selection.
 2. Moral Hazard
- How Adverse Selection Influences Financial Structure
 - How adverse selection effects the flow of capital.
 - Who can reduce this problem?
 1. Rating agencies
 2. Government Regulation to Increase Information
 3. Financial Intermediation
- The free-rider problem
 - If some use information for free.
 - Explains why banks play a more important role in financing the than securities

How Moral Hazard Affects the Choice Between Debt and Equity Contracts



Xi'an Jiaotong-Liverpool University
西交利物浦大學

Tools to Help Solve the Principal-Agent Problem

1. Production of Information: Monitoring
2. Government Regulation to Increase Information
3. Financial Intermediation (e.g., venture capital)

4. Debt Contracts

- Most debt contracts require the borrower to pay a fixed amount (interest) and borrower keep any profit above this amount
- Only cares about fixed return (interests+principle).
- Equity holder vs. Debt holder
 - Lower cost of state verification.
- More willing to become a debt holder if do not have enough information or low risk preference.